

出门票 5:

知识体系

Dynamical Oceanography ---To describe and understand oceanic processes via simple laws such as Newton's laws of motion

Difficulties: (1) Ocean is rotating, (2) Ocean is a continuum instead of discrete particles.

Long Range Goal ---> Prediction ---> Environmental Management

Concepts to be introduced:

(1) Geostrophy

(2) Geostrophy & Friction ----> Ekman layer & upwelling dynamics

(3) Geostrophy & Stratification ---> Gulf Stream structure & River plumes

(4) Vorticity

(5) Potential Vorticity

(6) Conservation of Potential Vorticity + Coastally Trapped Waves (Rossby, Continental Shelf & Kelvin Waves)

(7) Sverdrup Relationship + Davidson Current off California

(8) Western Boundary Currents ---> Gulf Stream and the Kuroshio

(9) El Niño and Southern Oscillation

学习了行星涡度 (Planetary Vorticity) 与位势涡度 (Potential Vorticity, 简称 PV) ,

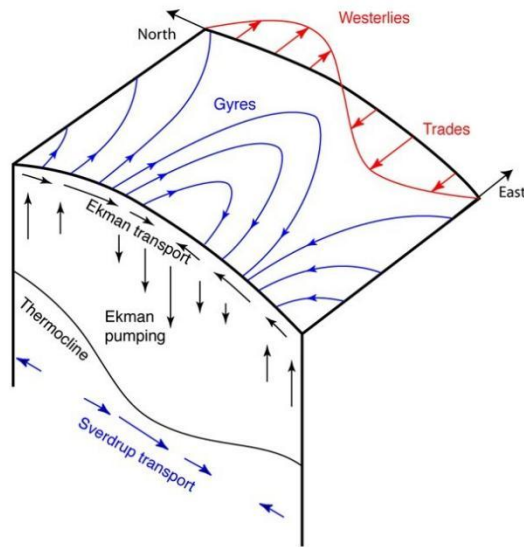
行星涡度 f 的大小与纬度有关, 即 $f = 2\Omega \cdot \sin(\phi)$, ϕ 是当地纬度。

无外力条件下, $PV = \frac{f + \zeta}{H} = \text{常量}$, 其中 H 是水柱厚度, ζ 是相对涡度。

应用: 向南移动时, f 变小, ζ 必须增大, 出现逆时针旋转

向北移动时, f 变大, ζ 必须减小, 出现顺时针旋转

Sverdrup Balance



- Wind causes Ekman transport and convergence
- Ekman pumping provides the squashing or stretching.
- The water columns must respond, either through change in relative vorticity or change in planetary vorticity (latitude). They do not spin up in place, but rather change latitude.

Squashing -> equatorward movement Stretching -> poleward

风驱海水辐合，埃克曼抽吸挤压 / 拉伸海水柱，海水柱为了保持位涡守恒，通过改变纬度（向赤道 / 极地运动）来平衡，最终形成了我们看到的大洋副热带涡旋，

重点：埃克曼抽吸（Ekman pumping）：海水辐合堆积后，会产生向下的下沉运动（图中向下的黑色箭头）；反之辐散区域会产生向上的上升运动。